Chapter 5. Fourier Transforms

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Overview

Results from Chapters 1–3 are cited as *Fact B.k* (see the *Background* notes). This chapter studies Fourier series of periodic functions and Fourier transforms of integrable functions, with Fubini's theorem and convolution as the key tools.

5.1 Fourier Series

Motivation. Is a periodic function the sum of its harmonics e^{inx}? Fejér's kernel gives a clean yes for Cesàro means and shows the coefficients determine the function.

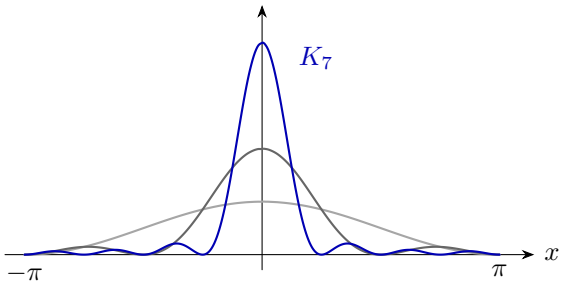


Figure. *Fejér kernels K_N ($N = 1, 3, 7$): nonnegative, integral 1, concentrating at 0, so $K_N * f \to f$.*

5.1.1 The circle $\mathbb{T}$

Let $\mathbb{T} = \mathbb{R}/\{2\pi n : n \in \mathbb{Z}\}$ with the distance from $\mathbb{R}$; 2π-periodic functions are functions on $\mathbb{T}$, and $C(\mathbb{T})$ consists of the continuous f on $[-\pi, \pi]$ with $f(-\pi) = f(\pi)$. As $\mathbb{T} = (-\pi, \pi]$, use Lebesgue measure normalized to mass 1:

$$\int_{\mathbb{T}} f = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx.$$

The convolution of §4.3 is defined on $L^1(\mathbb{T})$ in the same way, making $L^1(\mathbb{T})$ a Banach algebra: for $f, g : \mathbb{T} \to \mathbb{C}$, $t \in \mathbb{T}$,

$$(f * g)(t) = \int_{\mathbb{T}} f(s) g(t - s)\, ds.$$

If $f, g \in L^1(\mathbb{T})$, then as in §4.3, $(f * g)(t)$ is defined for almost every t, $f * g \in L^1(\mathbb{T})$,

$$\|f * g\|_1 \le \|f\|_1 \|g\|_1, \qquad f, g \in L^1(\mathbb{T}),$$

and the identities of Proposition 4.3.1 hold.

5.1.2 Fourier coefficients

Definition (Fourier coefficients, Fourier series)

For $f \in L^1(\mathbb{T})$ and $n = 0, \pm 1, \pm 2, \dots$ define

$$\widehat{f}(n) = \int_{\mathbb{T}} f(t) e^{-int}\, dt,$$

so that $|\widehat{f}(n)| \le \|f\|_1$. With

$$u_n(t) = e^{int}, \qquad t \in \mathbb{T}, \tag{1}$$

one checks directly that

$$u_n * f = \widehat{f}(n) u_n, \qquad n \in \mathbb{Z},\ f \in L^1(\mathbb{T}); \tag{2}$$

in particular $(u_n * f)(0) = \widehat{f}(n)$. Also $\widehat{u_n}(m) = 1$ if $m = n$ and 0 if $m \neq n$. The series

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n) u_n \tag{3}$$

is the **Fourier series** of $f \in L^1(\mathbb{T})$, with **Fourier coefficients** $\widehat{f}(n)$; without conditions on them, (3) need not define a function.

Indeed (2): $(u_n * f)(t) = \int_{\mathbb{T}} f(s) e^{in(t-s)}\, ds = e^{int}\widehat{f}(n)$.

Exercise 5.1.1

If $\widehat{f} \in \ell^1(\mathbb{Z})$, show that the series (3) converges uniformly to a continuous function.

Solution [Supplement]. $|\widehat{f}(n)u_n(t)| = |\widehat{f}(n)|$ and $\sum_n |\widehat{f}(n)| < \infty$, so by the M-test the partial sums (any $\sum_{-M}^{N}$) are uniformly Cauchy, with continuous uniform limit. □

Proposition 5.1.1

For $f, g \in L^1(\mathbb{T})$, constants $a, b \in \mathbb{C}$ and $n \in \mathbb{Z}$,

$$\widehat{af + bg}(n) = a\widehat{f}(n) + b\widehat{g}(n), \qquad \widehat{f * g}(n) = \widehat{f}(n)\,\widehat{g}(n). \tag{4}$$

Moreover for each $t \in \mathbb{T}$

$$\widehat{f_t}(n) = \widehat{f}(n) u_n(-t), \qquad n \in \mathbb{Z}, \tag{5}$$

where $f_t(s) = f(s - t)$.

Proof. The first identity is linearity (Fact B.10); for the second, by Fubini,

$$\begin{aligned}\widehat{f * g}(n) &= \int_{\mathbb{T}} (f * g)(t) e^{-int}\, dt = \int_{\mathbb{T}} e^{-int} \int_{\mathbb{T}} f(t - s) g(s)\, ds\, dt \\ &= \int_{\mathbb{T}} g(s) e^{-ins} \int_{\mathbb{T}} f(t - s) e^{-in(t-s)}\, dt\, ds = \int_{\mathbb{T}} g(s) e^{-ins}\, ds \int_{\mathbb{T}} f(t) e^{-int}\, dt = \widehat{g}(n)\widehat{f}(n).\end{aligned}$$

For (5),

$$\widehat{f_t}(n) = \int_{\mathbb{T}} f(s - t) e^{-ins}\, ds = \int_{\mathbb{T}} f(s) e^{-in(s+t)}\, ds = \widehat{f}(n) u_n(-t).$$

■

Proposition 5.1.2

If f is a 2π-periodic differentiable function with $f' \in L^1(\mathbb{T})$, then

$$\widehat{f'}(n) = in\widehat{f}(n), \qquad n \in \mathbb{Z}.$$

Proof. Integrating by parts (the boundary terms cancel by periodicity),

$$\widehat{f'}(n) = \int_{\mathbb{T}} f'(t) e^{-int}\, dt = in \int_{\mathbb{T}} f(t) e^{-int}\, dt = in\widehat{f}(n).$$

■

The next theorem is a variant of Fact B.21, proved as in Fact B.22.

Theorem 5.1.3

For each $p \in [1, \infty)$, the space $C(\mathbb{T})$ of continuous functions is dense in $L^p(\mathbb{T})$.

For a function f on $\mathbb{T}$ and $t \in \mathbb{T}$ define the translate as in Fact B.19:

$$f_t(s) = f(s - t), \qquad s \in \mathbb{T}.$$

The next theorem is proved like Fact B.23, with Theorem 5.1.3 in place of Fact B.22.

Theorem 5.1.4

For each $f \in L^p(\mathbb{T})$ $(1 \le p < \infty)$, the map

$$t \mapsto f_t : \mathbb{T} \to L^p(\mathbb{T})$$

is uniformly continuous.

Exercise 5.1.2

Prove Theorems 5.1.3 and 5.1.4.

Solution [Supplement]**.** *Theorem 5.1.3.* Simple functions are dense in $L^p(\mathbb{T})$ (Fact B.21), so approximate χ_E, $E \subset (-\pi, \pi)$ measurable. By regularity take compact K and open U with $K \subset E \subset U \subset (-\pi, \pi)$, $m(U \setminus K) < \epsilon$. Let

$$\varphi(x) = \frac{d(x, U^c)}{d(x, U^c) + d(x, K)},$$

continuous, $0 \le \varphi \le 1$, $\varphi = 1$ on K, $\varphi = 0$ off U; it vanishes near $\pm\pi$, so $\varphi \in C(\mathbb{T})$. As $|\chi_E - \varphi| \le \chi_{U \setminus K}$,

$$\|\chi_E - \varphi\|_p^p \le \frac{m(U \setminus K)}{2\pi} < \epsilon.$$

Taking linear combinations proves density.

Theorem 5.1.4. For $f \in L^p(\mathbb{T})$ take $g \in C(\mathbb{T})$ with $\|f - g\|_p < \epsilon$, uniformly continuous: $|g(u) - g(v)| < \epsilon$ for $d(u, v) < \delta$. By translation invariance $\|f_t - g_t\|_p = \|f - g\|_p$, and for $d(t, s) < \delta$, $\|g_t - g_s\|_p \le \|g_t - g_s\|_\infty \le \epsilon$ (mass 1), so

$$\|f_t - f_s\|_p \le \|f_t - g_t\|_p + \|g_t - g_s\|_p + \|g_s - f_s\|_p < 3\epsilon.$$

□

5.1.3 Approximate identities

The central theorem on Fourier series: by (7), $\langle h_n \rangle$ acts like an identity in the Banach algebra $L^1(\mathbb{T})$.

Theorem 5.1.5

If a sequence $\langle h_n \rangle$ satisfies

$$h_n \ge 0, \qquad \int_{\mathbb{T}} h_n = 1, \qquad \lim_{n \to \infty} \int_{\mathbb{T} \setminus [-\delta, \delta]} h_n = 0 \quad (0 < \delta < \pi), \tag{6}$$

(the third condition meaning: for *every* $\delta \in (0, \pi)$ the limit is 0), then for every $f \in L^1(\mathbb{T})$

$$\lim_{n \to \infty} \|f - h_n * f\|_1 = 0. \tag{7}$$

Proof. Let $\epsilon > 0$. By Theorem 5.1.4 there is $\delta > 0$ with

$$|s| < \delta \implies \|f - f_s\|_1 < \epsilon,$$

and then N with

$$n \geq N \implies \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds < \frac{\epsilon}{2\|f\|_1}.$$

For every $t \in \mathbb{T}$,

$$f(t) - (h_n * f)(t) = f(t)\int_{\mathbb{T}} h_n(s)\,ds - \int_{\mathbb{T}} h_n(s) f(t-s)\,ds = \int_{\mathbb{T}} h_n(s)\big[f(t) - f_s(t)\big]ds.$$

Hence for $n \geq N$, using Tonelli,

$$\begin{aligned}
\|f - h_n * f\|_1 &\leq \int_{\mathbb{T}}\int_{\mathbb{T}} h_n(s)|f(t) - f_s(t)|\,ds\,dt = \int_{\mathbb{T}} h_n(s)\int_{\mathbb{T}}|f(t) - f_s(t)|\,dt\,ds \\
&= \int_{\mathbb{T}} h_n(s)\|f - f_s\|_1\,ds \\
&= \int_{[-\delta,\delta]} h_n(s)\|f - f_s\|_1\,ds + \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\|f - f_s\|_1\,ds \\
&\leq \epsilon\int_{[-\delta,\delta]} h_n(s)\,ds + 2\|f\|_1\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds \leq \epsilon + \epsilon = 2\epsilon.
\end{aligned}$$

■

The same method gives the next theorem, more simply (no Tonelli–Fubini).

Theorem 5.1.6

If $\langle h_n\rangle$ satisfies (6), then $\lim_{n\to\infty}\|f - h_n * f\|_\infty = 0$ for every $f \in C(\mathbb{T})$.

Proof. Since f is uniformly continuous, choose $\delta > 0$ with

$$|s| < \delta \implies \|f - f_s\|_\infty < \epsilon,$$

and N with

$$n \geq N \implies \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds < \frac{\epsilon}{2\|f\|_\infty}.$$

Then for every $t \in \mathbb{T}$ and $n \geq N$,

$$\begin{aligned}
|f(t) - (h_n * f)(t)| &\leq \int_{\mathbb{T}} h_n(s)|f(t) - f_s(t)|\,ds = \Big(\int_{[-\delta,\delta]} + \int_{\mathbb{T}\setminus[-\delta,\delta]}\Big)h_n(s)|f(t) - f_s(t)|\,ds \\
&\leq \epsilon\int_{[-\delta,\delta]} h_n(s)\,ds + 2\|f\|_\infty\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds \leq \epsilon + \epsilon = 2\epsilon.
\end{aligned}$$

■

Exercise 5.1.3

In the proof of Theorem 5.1.6, show that there is $\delta > 0$ with $|s| < \delta \Rightarrow \|f - f_s\|_\infty < \epsilon$.

Solution [Supplement]. f is uniformly continuous on the compact $\mathbb{T}$: $|f(u) - f(v)| < \epsilon/2$ for $d(u,v) < \delta$. For $|s| < \delta$, $d(t, t-s) < \delta$, so $\|f - f_s\|_\infty \leq \epsilon/2 < \epsilon$. □

Exercise 5.1.4

Use Theorem 5.1.6 to prove Theorem 5.1.5.

Solution [Supplement]. Let $f \in L^1(\mathbb{T})$, $\epsilon > 0$, and choose $g \in C(\mathbb{T})$ with $\|f - g\|_1 < \epsilon$ (Theorem 5.1.3). Since $\mathbb{T}$ has total mass 1, $\|\cdot\|_1 \le \|\cdot\|_\infty$, and $\|h_n\|_1 = 1$, so

$$\begin{aligned}\|f - h_n * f\|_1 &\le \|f - g\|_1 + \|g - h_n * g\|_1 + \|h_n * (g - f)\|_1 \\ &\le \epsilon + \|g - h_n * g\|_\infty + \|h_n\|_1 \|g - f\|_1 < 2\epsilon + \|g - h_n * g\|_\infty.\end{aligned}$$

By Theorem 5.1.6 the last term tends to 0, so $\limsup_n \|f - h_n * f\|_1 \le 2\epsilon$. □

5.1.4 The Fejér kernel

Sequences satisfying (6) are easy to find, e.g. $\langle n\pi\chi_{[-1/n,1/n]} : n = 1, 2, \dots \rangle$. A more important example: for $n = 1, 2, \dots$ put

$$K_n = \sum_{k=-n}^{n} \Big(1 - \frac{|k|}{n+1}\Big) u_k. \tag{8}$$

$\langle K_n \rangle$ is the **Fejér kernel**.

Exercise 5.1.5

Show that $\lim_{n\to\infty} \widehat{K_n}(k) = 1$ for every $k \in \mathbb{Z}$.

Solution [Supplement]. Since $\widehat{u_j}(k) = \delta_{jk}$, $\widehat{K_n}(k) = 1 - \frac{|k|}{n+1}$ for $|k| \le n$ and 0 for $|k| > n$. For fixed k and $n \ge |k|$ this is $1 - |k|/(n+1) \to 1$. □

Since $\int_{\mathbb{T}} u_n = 0$ for $n \in \mathbb{Z} \setminus \{0\}$, the second condition $\int_{\mathbb{T}} K_n = 1$ is immediate. Expanding the left side shows

$$\Big(-\frac{1}{4}u_{-1} + \frac{1}{2} - \frac{1}{4}u_1\Big) K_n(t) = \frac{1}{n+1}\Big(-\frac{1}{4}u_{-(n+1)} + \frac{1}{2} - \frac{1}{4}u_{n+1}\Big),$$

from which

$$K_n(t) = \frac{1}{n+1} \frac{\sin^2\big(\frac{n+1}{2}t\big)}{\sin^2\big(\frac{1}{2}t\big)}, \qquad n = 0, 1, 2, \dots \tag{9}$$

Exercise 5.1.6

Prove (9).

Solution [Supplement]. For $0 \le j, l \le n$, the number of pairs with $j - l = k$ is $n + 1 - |k|$ ($|k| \le n$). Hence

$$\Big|\sum_{j=0}^{n} e^{ijt}\Big|^2 = \sum_{j,l=0}^{n} e^{i(j-l)t} = \sum_{k=-n}^{n} (n + 1 - |k|) e^{ikt} = (n+1) K_n(t).$$

For $t \neq 0$,

$$\sum_{j=0}^{n} e^{ijt} = \frac{e^{i(n+1)t} - 1}{e^{it} - 1} = \frac{e^{i(n+1)t/2}}{e^{it/2}} \cdot \frac{\sin\frac{(n+1)t}{2}}{\sin\frac{t}{2}},$$

whose squared modulus is $\sin^2\frac{(n+1)t}{2} / \sin^2\frac{t}{2}$: this is (9). (Equivalently, $-\frac{1}{4}u_{-1} + \frac{1}{2} - \frac{1}{4}u_1 = \sin^2\frac{t}{2}$, and similarly on the right.) □

Since $t \mapsto \sin^2(\frac{1}{2}t)$ is increasing on $[0, \pi]$ and K_n is even, for every $\delta > 0$

$$0 \le K_n(t) \le \frac{1}{n+1} \frac{1}{\sin^2(\frac{1}{2}\delta)}, \qquad t \in \mathbb{T} \setminus [-\delta, \delta],$$

so $\langle K_n \rangle$ satisfies all of (6).

Exercise 5.1.7

Sketch the graph of K_n for $n = 1, 2, 3, 5, 10$.

Solution [Supplement]. $K_1(t) = 1 + \cos t$, $K_2(t) = 1 + \frac{4}{3}\cos t + \frac{2}{3}\cos 2t$; in general K_n is even, nonnegative, with maximum $K_n(0) = n + 1$, zeros at $t = \frac{2\pi j}{n+1}$, humps of height $\leq \frac{1}{(n+1)\sin^2(t/2)}$ between them, and average 1. For $n = 1, 2, 3, 5, 10$ the peak height is $2, 3, 4, 6, 11$ and its width $\frac{4\pi}{n+1}$ shrinks: the mass concentrates at 0. □

5.1.5 Consequences

Each K_n is a finite linear combination of the u_n: a **trigonometric polynomial** (space $T(\mathbb{T})$). $P = \sum_{n=-N}^{N} a_n u_n$ has $\widehat{P}(n) = a_n$.

By (2), $K_n * f$ is a trigonometric polynomial for every $f \in L^1(\mathbb{T})$, and by Theorem 5.1.5, $K_n * f \to f$ in L^1. Hence:

Theorem 5.1.7

$T(\mathbb{T})$ is dense in $L^1(\mathbb{T})$.

If $\widehat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $K_n * f = 0$ by (2), so $\|f\|_1 = \lim \|f - K_n * f\|_1 = 0$:

Theorem 5.1.8 (uniqueness)

If $f \in L^1(\mathbb{T})$ and $\widehat{f} = 0$, then $f = 0$ almost everywhere.

For $a \in \ell^1(\mathbb{Z})$, $\sum_k |a_k u_k(t)| = \sum_k |a_k| < \infty$, so $g = \sum_k a_k u_k$ is defined, and by Fubini for counting measure (Exercise 4.2.3)

$$\widehat{g}(n) = \int_{\mathbb{T}} \Big(\sum_{k \in \mathbb{Z}} a_k u_{k-n} \Big) = \sum_{k \in \mathbb{Z}} a_k \int_{\mathbb{T}} u_{k-n} = a_n.$$

In particular, if $f \in L^1(\mathbb{T})$ and $\widehat{f} \in \ell^1(\mathbb{Z})$, then $g = \sum_k \widehat{f}(k) u_k$ and f have the same Fourier coefficients, so Theorem 5.1.8 gives:

Corollary 5.1.9

If $f \in L^1(\mathbb{T})$ and $\widehat{f} \in \ell^1(\mathbb{Z})$, then

$$f(t) = \sum_{n \in \mathbb{Z}} \widehat{f}(n) e^{int}$$

for almost every $t \in \mathbb{T}$.

On the other hand, given $\epsilon > 0$ take $P \in T(\mathbb{T})$ with $\|f - P\|_1 < \epsilon$ and let $N = \max\{|n| : \widehat{P}(n) \neq 0\}$. For $|n| > N$,

$$|\widehat{f}(n)| = |\widehat{f}(n) - \widehat{P}(n)| \leq \|f - P\|_1 < \epsilon.$$

This proves:

Theorem 5.1.10 (Riemann–Lebesgue)

$\lim_{n \to \pm\infty} \widehat{f}(n) = 0$ for every $f \in L^1(\mathbb{T})$.

Exercise 5.1.8

If a 2π-periodic function $f : \mathbb{R} \to \mathbb{C}$ is twice differentiable and $f'' \in L^1(\mathbb{T})$, show that $\widehat{f} \in \ell^1(\mathbb{Z})$.

Solution [Supplement]. f' is differentiable, hence continuous and so bounded on $\mathbb{T}$; thus $f' \in L^1(\mathbb{T})$. Applying Proposition 5.1.2 to f and to f',

$$\widehat{f''}(n) = in\,\widehat{f'}(n) = (in)^2\widehat{f}(n) = -n^2\widehat{f}(n).$$

Hence $|\widehat{f}(n)| = |\widehat{f''}(n)|/n^2 \le \|f''\|_1/n^2$ for $n \neq 0$, and $\sum_n |\widehat{f}(n)| \le |\widehat{f}(0)| + \|f''\|_1 \sum_{n\neq 0} n^{-2} < \infty$. □

Let $c_0(\mathbb{Z})$ be the vector space of functions $a : \mathbb{Z} \to \mathbb{C}$ with $\lim_{n\to\pm\infty} a(n) = 0$, with the norm of $\ell^\infty(\mathbb{Z})$.

Exercise 5.1.9

Show that the normed space $c_0(\mathbb{Z})$ is a Banach space.

Solution [Supplement]. As $\ell^\infty(\mathbb{Z})$ is complete, show $c_0(\mathbb{Z})$ is closed. If $\|a_k - a\|_\infty < \epsilon$ and $|a_k(n)| < \epsilon$ for $|n| > N$, then $|a(n)| < 2\epsilon$ for $|n| > N$. □

Summarizing:

Theorem 5.1.11

The map $f \mapsto \widehat{f}$ is a norm-decreasing, injective linear map from the Banach algebra $L^1(\mathbb{T})$ into $c_0(\mathbb{Z})$.

Proof. Linearity is (4); $\|\widehat{f}\|_\infty \le \|f\|_1$ is the estimate $|\widehat{f}(n)| \le \|f\|_1$; $\widehat{f} \in c_0(\mathbb{Z})$ is Theorem 5.1.10; injectivity is Theorem 5.1.8. ■

With $(ab)(n) = a(n)b(n)$, $c_0(\mathbb{Z})$ is a Banach algebra, and the last identity of (4) says $f \mapsto \widehat{f}$ preserves multiplication: a **homomorphism** of Banach algebras.

Remark ($L^1(\mathbb{T})$ has no identity)

If the Banach algebra $L^1(\mathbb{T})$ had an identity e, then for every $n \in \mathbb{Z}$

$$\widehat{e}(n) = \widehat{e}(n)\widehat{u_n}(n) = \widehat{e * u_n}(n) = \widehat{u_n}(n) = 1,$$

contradicting $\widehat{e} \in c_0(\mathbb{Z})$. So $L^1(\mathbb{T})$ has no identity; Theorem 5.1.5 supplies an *approximate identity*.

Exercise 5.1.10

If $\langle h_n \rangle$ satisfies (6), show that $\|a\widehat{h_n} - a\|_\infty \to 0$ for every $a \in c_0(\mathbb{Z})$.

Solution [Supplement]. First, $|\widehat{h_n}(m)| \le \|h_n\|_1 = 1$. Next, for fixed m and $0 < \delta < \pi$,

$$|1 - \widehat{h_n}(m)| = \Big|\int_{\mathbb{T}} h_n(s)(1 - e^{-ims})\,ds\Big| \le \int_{[-\delta,\delta]} h_n(s)|ms|\,ds + 2\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n \le |m|\delta + 2\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n,$$

so $\limsup_n |1 - \widehat{h_n}(m)| \le |m|\delta$ for every δ, i.e. $\widehat{h_n}(m) \to 1$. Now let $\epsilon > 0$ and choose M with $|a(m)| < \epsilon$ for $|m| > M$. Then

$$\|a\widehat{h_n} - a\|_\infty \le \max_{|m|\le M} |a(m)||\widehat{h_n}(m) - 1| + \sup_{|m|>M} |a(m)| \cdot 2,$$

where the first term tends to 0 (finitely many m) and the second is at most 2ϵ. □

Finally, consider some consequences of Theorem 5.1.6. With the Fejér kernel, $K_n * f \in T(\mathbb{T})$ and $K_n * f \to f$ uniformly for $f \in C(\mathbb{T})$:

Corollary 5.1.12

$T(\mathbb{T})$ is dense in $C(\mathbb{T})$ (in the uniform norm).

If $f \in L^p(\mathbb{T})$ $(1 \le p < \infty)$, Theorem 5.1.3 gives $g \in C(\mathbb{T})$ with $\|f - g\|_p < \epsilon$, and Corollary 5.1.12 gives $P \in T(\mathbb{T})$ with $\|g - P\|_\infty < \epsilon$. Then

$$\|f - P\|_p \le \|f - g\|_p + \|g - P\|_p < \epsilon + \|g - P\|_\infty < 2\epsilon.$$

Corollary 5.1.13

$T(\mathbb{T})$ is dense in $L^p(\mathbb{T})$ for $1 \le p < \infty$.

Corollary 5.1.12 implies the Weierstrass polynomial approximation theorem on bounded closed intervals.

Theorem 5.1.14 (Weierstrass)

The space of polynomial functions on $[0,1]$ is dense in $C[0,1]$.

Proof. Extend a continuous $f : [0,1] \to \mathbb{C}$ to a continuous periodic $g : \mathbb{T} \to \mathbb{C}$. By Corollary 5.1.12 choose $\{a_k \in \mathbb{C} : k = 0, \pm 1, \dots, \pm n\}$ with

$$\Big| f(t) - \sum_{k=-n}^{n} a_k u_k(t) \Big| < \epsilon, \qquad t \in [0,1].$$

Since the Taylor series of the exponential converges uniformly on $[0,1]$, for each k there is $M(k) \in \mathbb{N}$ with

$$\Big| \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!} - u_k(t) \Big| \le \frac{\epsilon}{(2n+1)|a_k| + 1}, \qquad t \in [0,1].$$

Put

$$P(t) = \sum_{k=-n}^{n} a_k \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!}, \qquad t \in [0,1].$$

Then $|f(t) - P(t)| < 2\epsilon$ (Exercise 5.1.11). ■

Exercise 5.1.11

Complete the proof of Theorem 5.1.14.

Solution [Supplement]. *Extension.* Define g on $[-\pi, \pi]$: $g = f$ on $[0,1]$, linear from $f(1)$ to 0 on $[1,2]$, 0 on $[2,\pi] \cup [-\pi,-1]$, linear from 0 to $f(0)$ on $[-1,0]$. Then $g \in C(\mathbb{T})$.

Estimate. P is a polynomial in t. For $t \in [0,1]$,

$$\begin{aligned} |f(t) - P(t)| &\le \Big| f(t) - \sum_{k=-n}^{n} a_k u_k(t) \Big| + \sum_{k=-n}^{n} |a_k| \Big| u_k(t) - \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!} \Big| \\ &< \epsilon + \sum_{k=-n}^{n} \frac{|a_k|\epsilon}{(2n+1)|a_k| + 1} < \epsilon + (2n+1) \cdot \frac{\epsilon}{2n+1} = 2\epsilon. \end{aligned}$$

(If f is real-valued, $\operatorname{Re} P$ is a real polynomial with $|f - \operatorname{Re} P| \le |f - P|$.) □

5.2 The Fourier Integral

Motivation. On $\mathbb{R}$ sums over frequencies become integrals: continuity and decay of $\hat{f}$, inversion via Gauss and Fejér kernels, differentiation as multiplication.

5.2.1 Definition and continuity

Definition (Fourier transform)

For $f \in L^1(\mathbb{R})$ and $\alpha \in \mathbb{R}$ define

$$\widehat{f}(\alpha) = \int_{\mathbb{R}} f(x) e^{-i\alpha x}\, dx.$$

$\widehat{f}$ is the **Fourier transform** of f.

Clearly $|\widehat{f}(\alpha)| \le \|f\|_1$ for all $\alpha \in \mathbb{R}$. If $\alpha_n \to \alpha$, then

$$|\widehat{f}(\alpha_n) - \widehat{f}(\alpha)| \le \int_{\mathbb{R}} |f(x)||e^{-i\alpha_n x} - e^{-i\alpha x}|\, dx,$$

and $|e^{-i\alpha_n x} - e^{-i\alpha x}| \le 2$, $e^{-i\alpha_n x} \to e^{-i\alpha x}$, so dominated convergence gives $\widehat{f}(\alpha_n) \to \widehat{f}(\alpha)$: $\widehat{f}$ *is bounded and continuous.* Writing

$$u_\alpha(x) = e^{i\alpha x}, \qquad x \in \mathbb{R},$$

one checks immediately that

$$u_\alpha * f = \widehat{f}(\alpha) u_\alpha, \qquad \alpha \in \mathbb{R}.$$

Example 1

For $f(x) = e^{-|x|/a}$ $(a > 0)$,

$$\widehat{f}(\alpha) = \int_{\mathbb{R}} e^{-|x|/a} e^{-i\alpha x}\, dx = \int_{-\infty}^{0} e^{(1/a - i\alpha)x}\, dx + \int_0^{\infty} e^{(-1/a - i\alpha)x}\, dx = \frac{2a}{1 + a^2\alpha^2}.$$

For $g = \chi_{[a,b]}$,

$$\widehat{g}(\alpha) = \int_{-\infty}^{\infty} \chi_{[a,b]} e^{-i\alpha x}\, dx = \frac{e^{-i\alpha a} - e^{-i\alpha b}}{i\alpha},$$

and in particular

$$\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin \alpha a}{\alpha}, \qquad a > 0.$$

Exercise 5.2.1

In Example 1, describe how the graphs of f and $\widehat{f}$ change as $a \to \infty$. Do the same for $\widehat{\chi_{[-a,a]}}$.

Solution [Supplement]. As $a \to \infty$, $e^{-|x|/a}$ spreads out to 1, while $\widehat{f}(\alpha) = \frac{2a}{1+a^2\alpha^2}$ (peak $2a$, half-width $1/a$, $\int \widehat{f} = 2\pi f(0) = 2\pi$) concentrates at 0 like $2\pi\delta_0$. Likewise $\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin a\alpha}{\alpha}$ has peak $2a$, zeros at $k\pi/a$ $(k \neq 0)$ and oscillations decaying like $2/|\alpha|$: wide functions have narrow transforms. (It is not in $L^1(\mathbb{R})$.) □

The basic properties of Proposition 5.1.1 hold for Fourier integrals as well.

Proposition 5.2.1

For $f, g \in L^1(\mathbb{R})$ and constants $a, b \in \mathbb{C}$,

$$\widehat{af + bg}(\alpha) = a\widehat{f}(\alpha) + b\widehat{g}(\alpha), \qquad \widehat{f * g}(\alpha) = \widehat{f}(\alpha)\,\widehat{g}(\alpha), \qquad \alpha \in \mathbb{R}.$$

Moreover for each $\alpha \in \mathbb{R}$

$$\widehat{fu_\alpha} = (\widehat{f})_\alpha, \qquad \widehat{f_x}(\alpha) = \widehat{f}(\alpha)u_\alpha(-x), \quad x \in \mathbb{R}.$$

Proof. For example, for every $\beta \in \mathbb{R}$

$$\widehat{fu_\alpha}(\beta) = \int_{\mathbb{R}} f(x)u_\alpha(x)e^{-i\beta x}\,dx = \int_{\mathbb{R}} f(x)e^{-i(\beta-\alpha)x}\,dx = \widehat{f}(\beta - \alpha),$$

so $\widehat{fu_\alpha} = (\widehat{f})_\alpha$. The rest is Exercise 5.2.2. ■

Exercise 5.2.2

Complete the proof of Proposition 5.2.1.

Solution [Supplement]**.** Linearity is clear. $(x, y) \mapsto f(x - y)g(y)e^{-i\alpha x}$ is integrable on $\mathbb{R}^2$ (with integral of the modulus $\|f\|_1\|g\|_1$), so by Fubini

$$\widehat{f * g}(\alpha) = \int_{\mathbb{R}} g(y)e^{-i\alpha y} \int_{\mathbb{R}} f(x - y)e^{-i\alpha(x-y)}\,dx\,dy = \widehat{g}(\alpha)\widehat{f}(\alpha).$$

For translation, $\widehat{f_x}(\alpha) = \int_{\mathbb{R}} f(y - x)e^{-i\alpha y}\,dy = \int_{\mathbb{R}} f(y)e^{-i\alpha(y+x)}\,dy = \widehat{f}(\alpha)e^{-i\alpha x} = \widehat{f}(\alpha)u_\alpha(-x)$. □

5.2.2 Differentiation

Proposition 5.2.2

Let $f \in L^1(\mathbb{R})$.

(a) Let $g(x) = -ixf(x)$. If $g \in L^1(\mathbb{R})$, then $\widehat{f}$ is differentiable and

$$\widehat{f}\,'(\alpha) = \widehat{g}(\alpha), \qquad \alpha \in \mathbb{R}.$$

(b) If f is differentiable and $f' \in L^1(\mathbb{R})$, then

$$\widehat{f'}(\alpha) = i\alpha\widehat{f}(\alpha), \qquad \alpha \in \mathbb{R}.$$

Proof. (a) The difference quotient is

$$\frac{\widehat{f}(\alpha + h) - \widehat{f}(\alpha)}{h} = \int_{\mathbb{R}} f(x)e^{-i\alpha x}\frac{e^{-ihx} - 1}{h}\,dx,$$

and

$$\left|f(x)e^{-i\alpha x}\frac{e^{-ihx} - 1}{h}\right| \le |xf(x)|.$$

Since $x \mapsto |xf(x)|$ is integrable and $\lim_{h\to 0}\frac{e^{-ihx}-1}{h} = -ix$, the Lebesgue convergence theorem gives, for every real sequence $h_n \to 0$,

$$\lim_{n\to\infty}\frac{\widehat{f}(\alpha + h_n) - \widehat{f}(\alpha)}{h_n} = \widehat{g}(\alpha).$$

(b) By parts as in Proposition 5.1.2: $\widehat{f'}(\alpha) = i\alpha\int_{\mathbb{R}} f(x)e^{-i\alpha x}\,dx$, the boundary terms vanishing as $f(\pm\infty)$ exist ($f' \in L^1$) and are 0 ($f \in L^1$). ■

5.2.3 Riemann–Lebesgue lemma

As for Fourier series, $\lim_{\alpha\to\pm\infty}\widehat{f}(\alpha)=0$ for $f\in L^1(\mathbb{R})$. Using $e^{\pi i}=-1$,

$$\widehat{f}(\alpha)=-\int_{\mathbb{R}} f(x)e^{-i\alpha(x+\pi/\alpha)}\,dx=-\int_{\mathbb{R}} f\Big(x-\frac{\pi}{\alpha}\Big)e^{-i\alpha x}\,dx,$$

and hence

$$2\widehat{f}(\alpha)=\int_{\mathbb{R}}[f(x)-f_{\pi/\alpha}(x)]e^{-i\alpha x}\,dx.$$

By Fact B.23,

$$0\le\lim_{\alpha\to\pm\infty}|\widehat{f}(\alpha)|\le\frac{1}{2}\lim_{\alpha\to\pm\infty}\|f-f_{\pi/\alpha}\|_1=0.$$

Since $\widehat{f}$ is already known to be continuous:

Theorem 5.2.3 (Riemann–Lebesgue)

$\widehat{f}\in C_0(\mathbb{R})$ for every $f\in L^1(\mathbb{R})$.

5.2.4 Approximate identities on $\mathbb{R}$

Using Fact B.23 instead of Theorem 5.1.4 in the proof of Theorem 5.1.5 yields:

Theorem 5.2.4

If a family $\{h_\lambda:\lambda\in(0,\infty)\}$ satisfies

$$h_\lambda\ge 0,\qquad\int_{\mathbb{R}}h_\lambda=1,\qquad\lim_{\lambda\to\infty}\int_{\mathbb{R}\setminus[-\delta,\delta]}h_\lambda=0\quad(\delta>0),\tag{10}$$

then:

(a) $\lim_{\lambda\to\infty}\|f-h_\lambda*f\|_1=0$ for every $f\in L^1(\mathbb{R})$;

(b) $\lim_{\lambda\to\infty}\|f-h_\lambda*f\|_\infty=0$ for every bounded uniformly continuous $f:\mathbb{R}\to\mathbb{C}$.

Exercise 5.2.3

Prove Theorem 5.2.4.

Solution [Supplement]. (a) Let $\epsilon>0$. By Fact B.23 choose $\delta>0$ with $|s|<\delta\Rightarrow\|f-f_s\|_1<\epsilon$, and Λ with $\int_{\mathbb{R}\setminus[-\delta,\delta]}h_\lambda<\epsilon/(2\|f\|_1)$ for $\lambda\ge\Lambda$. Since $f(t)-(h_\lambda*f)(t)=\int_{\mathbb{R}}h_\lambda(s)[f(t)-f_s(t)]\,ds$, Tonelli gives for $\lambda\ge\Lambda$

$$\|f-h_\lambda*f\|_1\le\int_{\mathbb{R}}h_\lambda(s)\|f-f_s\|_1\,ds\le\epsilon\int_{[-\delta,\delta]}h_\lambda+2\|f\|_1\int_{\mathbb{R}\setminus[-\delta,\delta]}h_\lambda<2\epsilon.$$

(b) Choose δ with $|s|<\delta\Rightarrow\|f-f_s\|_\infty<\epsilon$ (uniform continuity) and Λ with $\int_{\mathbb{R}\setminus[-\delta,\delta]}h_\lambda<\epsilon/(2\|f\|_\infty)$ for $\lambda\ge\Lambda$. Then for every t,

$$|f(t)-(h_\lambda*f)(t)|\le\int_{\mathbb{R}}h_\lambda(s)|f(t)-f_s(t)|\,ds\le\epsilon+2\|f\|_\infty\cdot\frac{\epsilon}{2\|f\|_\infty}=2\epsilon.$$

□

If $h:\mathbb{R}\to\mathbb{C}$ satisfies

$$h\ge 0,\qquad\int_{\mathbb{R}}h=1,\tag{11}$$

put

$$h_\lambda(x) = \lambda h(\lambda x), \qquad x \in \mathbb{R},\ \lambda > 0. \tag{12}$$

Then $\{h_\lambda : \lambda > 0\}$ satisfies (10): $h_\lambda \geq 0$ and $\int h_\lambda = 1$ are clear, and for fixed $\delta > 0$, as $\lambda \to \infty$,

$$\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda(x)\,dx = \int_{\mathbb{R}\setminus[-\lambda\delta,\lambda\delta]} h(x)\,dx \to 0.$$

Exercise 5.2.4

Show that $\widehat{h_\lambda}(\alpha) = \widehat{h}(\frac{\alpha}{\lambda})$ for every $\alpha \in \mathbb{R}$.

Solution [Supplement]**.** Substituting $y = \lambda x$: $\widehat{h_\lambda}(\alpha) = \int_{\mathbb{R}} \lambda h(\lambda x) e^{-i\alpha x}\,dx = \int_{\mathbb{R}} h(y) e^{-i(\alpha/\lambda)y}\,dy = \widehat{h}(\alpha/\lambda)$. □

The case where h satisfying (11) has the form

$$h(x) = \frac{1}{2\pi}\int_{\mathbb{R}} p(\alpha) u_\alpha(x)\,d\alpha \tag{13}$$

is very useful. Here $p : \mathbb{R} \to \mathbb{C}$ is assumed to satisfy

$$0 \leq p \leq 1, \qquad p(0) = \lim_{\alpha\to 0} p(\alpha) = 1, \qquad p \in L^1(\mathbb{R}). \tag{14}$$

A simple example satisfying all requirements:

$$H(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-|\alpha|} e^{i\alpha x}\,d\alpha = \frac{1}{\pi}\frac{1}{1+x^2}, \tag{15}$$

and then

$$H_\lambda(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-|\alpha/\lambda|} e^{i\alpha x}\,d\alpha = \frac{1}{2\pi}\frac{2\lambda}{1+\lambda^2 x^2}.$$

$\{H_\lambda : \lambda > 0\}$ is the **Poisson kernel**.

5.2.5 Smooth functions are dense

Let $C_c^n(\mathbb{R})$ be the $f \in C_c(\mathbb{R})$ with continuous $f^{(n)}$, and $C_c^\infty(\mathbb{R}) = \bigcap_n C_c^n(\mathbb{R})$. If $h \in C_c^\infty(\mathbb{R})$ satisfies (11), then $h_\lambda \in C_c^\infty(\mathbb{R})$ and, by Proposition 4.3.2 repeatedly, $h_\lambda * f \in C_c^\infty(\mathbb{R})$ for $f \in C_c(\mathbb{R})$. By Theorem 5.2.4

$$\lim_{\lambda\to\infty} \|f - h_\lambda * f\|_1 = \lim_{\lambda\to\infty} \|f - h_\lambda * f\|_\infty = 0.$$

Since $C_c(\mathbb{R})$ is dense in $L^1(\mathbb{R})$ and in $C_0(\mathbb{R})$:

Proposition 5.2.5

$C_c^\infty(\mathbb{R})$ is dense in the normed spaces $L^1(\mathbb{R})$ and $C_0(\mathbb{R})$.

Exercise 5.2.5

Give an example of $h \in C_c^\infty(\mathbb{R})$ satisfying (11).

Solution [Supplement]**.** Let $\varphi(x) = \exp(-\frac{1}{1-x^2})$ for $|x| < 1$, 0 otherwise. $\psi(t) = e^{-1/t}$ $(t > 0)$, 0 $(t \leq 0)$ is C^∞ (derivatives $P(1/t)e^{-1/t} \to 0$ as $t \downarrow 0$), so $\varphi = \psi(1-x^2) \in C_c^\infty(\mathbb{R})$, $\varphi \geq 0$, $\int \varphi > 0$. Take $h = \varphi / \int_{\mathbb{R}} \varphi$. □

5.2.6 The inversion formula

For h given by (13) and $\{h_\lambda\}$ by (12), compute $h_\lambda * f$ for $f \in L^1(\mathbb{R})$ using Fubini:

$$\begin{aligned}(h_\lambda * f)(x) &= \int_{\mathbb{R}} f(x-y)\frac{\lambda}{2\pi}\int_{\mathbb{R}} p(\alpha)e^{i\alpha\lambda y}\,d\alpha\,dy = \frac{1}{2\pi}\int_{\mathbb{R}} f(x-y)\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})e^{i\alpha y}\,d\alpha\,dy \\ &= \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\int_{\mathbb{R}} f(x-y)e^{i\alpha y}\,dy\,d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\int_{\mathbb{R}} f(y)e^{i\alpha(x-y)}\,dy\,d\alpha \\ &= \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\widehat{f}(\alpha)e^{i\alpha x}\,d\alpha.\end{aligned}$$

(Fubini applies since $\iint |f(x-y)||p(\alpha/\lambda)|\,d\alpha\,dy = \lambda\|f\|_1\|p\|_1 < \infty$.)

From now on assume p, h satisfy (14), (11) and (13). Then

$$(h_\lambda * f)(x) = \frac{1}{2\pi}\int_{\mathbb{R}} p\Big(\frac{\alpha}{\lambda}\Big)\widehat{f}(\alpha)e^{i\alpha x}\,d\alpha.$$

Since

$$\Big|p\Big(\frac{\alpha}{\lambda}\Big)\widehat{f}(\alpha)e^{i\alpha x}\Big| \le |\widehat{f}(\alpha)|, \qquad \lim_{\lambda\to\infty} p\Big(\frac{\alpha}{\lambda}\Big) = 1,$$

if $\widehat{f} \in L^1(\mathbb{R})$ the Lebesgue convergence theorem gives

$$\lim_{\lambda\to\infty}(h_\lambda * f)(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\alpha)e^{i\alpha x}\,d\alpha, \qquad x \in \mathbb{R}.$$

On the other hand $\lim_{\lambda\to\infty}\|h_\lambda * f - f\|_1 = 0$ by Theorem 5.2.4, so by Fact B.18 there is a sequence $\lambda_n \to \infty$ with

$$\lim_{n\to\infty}(h_{\lambda_n} * f)(x) = f(x)$$

for almost every x. This proves the analogue of Corollary 5.1.9 (with, e.g., $p(\alpha) = e^{-|\alpha|}$ and $h = H$ from (15)):

Theorem 5.2.6 (inversion formula)

If $f \in L^1(\mathbb{R})$ and $\widehat{f} \in L^1(\mathbb{R})$, then

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\alpha)e^{i\alpha x}\,d\alpha \tag{16}$$

for almost every x.

The right side of (16) is continuous in x, so for continuous f, (16) holds everywhere.

Example 2

Applying the inversion formula to Example 1 (with $a = 1$),

$$e^{-|x|} = \int_{-\infty}^{\infty}\frac{1}{\pi(1+\alpha^2)}e^{i\alpha x}\,d\alpha = \int_{-\infty}^{\infty}\frac{\cos\alpha x}{\pi(1+\alpha^2)}\,d\alpha + i\int_{-\infty}^{\infty}\frac{\sin\alpha x}{\pi(1+\alpha^2)}\,d\alpha.$$

Since the left side is real (and the cosine integrand is even),

$$\int_0^{\infty}\frac{\cos\alpha x}{1+\alpha^2}\,d\alpha = \frac{\pi}{2}e^{-|x|}.$$

If $\widehat{f} = 0$, then $\widehat{f} \in L^1$ and (16) gives $f = 0$ almost everywhere, the analogue of Theorem 5.1.8:

Theorem 5.2.7 (uniqueness)

If $f \in L^1(\mathbb{R})$ and $\widehat{f} = 0$, then $f = 0$ almost everywhere.

5.2.7 Fejér and Gauss kernels

Two more p satisfying (14), (11), (13). With the Fejér kernel (8), for $\lambda > 0$ put

$$K_\lambda(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \max\Big\{1 - \frac{|\alpha|}{\lambda}, 0\Big\} u_\alpha(x)\, d\alpha = \frac{1}{2\pi}\int_{-\lambda}^{\lambda} \Big(1 - \frac{|\alpha|}{\lambda}\Big) e^{i\alpha x}\, d\alpha.$$

Then

$$\lambda K_1(\lambda x) = \frac{1}{2\pi}\int_{-1}^{1} (1 - |\alpha|) e^{i\alpha\lambda x} \lambda\, d\alpha = K_\lambda(x),$$

so with $\lambda = 1$ in mind define

$$K(x) = \frac{1}{2\pi}\int_{-1}^{1} (1 - |\alpha|) e^{i\alpha x}\, d\alpha.$$

A direct computation (integration by parts) gives

$$K_\lambda(x) = \frac{1}{\pi}\int_0^\lambda \Big(1 - \frac{\alpha}{\lambda}\Big)\cos\alpha x\, d\alpha = \frac{1}{\pi\lambda x^2}(1 - \cos\lambda x) = \frac{1}{2\pi\lambda}\,\frac{\sin^2(\frac{\lambda x}{2})}{(\frac{x}{2})^2},$$

so $K \geq 0$. As $\alpha \mapsto -\alpha$ leaves K_λ unchanged, $K_\lambda = \frac{1}{2\pi}\widehat{f_\lambda}$ with $f_\lambda(\alpha) = \max\{1 - \frac{|\alpha|}{\lambda}, 0\}$; both are integrable ($\widehat{f_\lambda} = O(x^{-2})$), so inversion gives

$$\int_{-\infty}^{\infty} K_\lambda(x) e^{i\alpha x}\, dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f_\lambda}(x) e^{i\alpha x}\, dx = f_\lambda(\alpha) = \max\Big\{1 - \frac{|\alpha|}{\lambda}, 0\Big\}.$$

Since f_λ is continuous, this holds for every $\alpha \in \mathbb{R}$; in particular for $\lambda = 1$, $\alpha = 0$,

$$\int_{-\infty}^{\infty} K(x)\, dx = 1.$$

The family $\{K_\lambda : \lambda > 0\}$ obtained this way is also called the **Fejér kernel**.

Finally let

$$G(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\alpha^2/2} u_\alpha(x)\, d\alpha.$$

By Fact B.11 (differentiation under the integral) and integration by parts,

$$G'(x) = \frac{1}{2\pi}\int_{\mathbb{R}} i\alpha e^{-\alpha^2/2} e^{i\alpha x}\, d\alpha = -xG(x)$$

(indeed $\int i\alpha e^{-\alpha^2/2} e^{i\alpha x}\, d\alpha = -i\int (e^{-\alpha^2/2})' e^{i\alpha x}\, d\alpha = i \cdot ix \int e^{-\alpha^2/2} e^{i\alpha x}\, d\alpha$). Since $G(0) = \frac{1}{2\pi}\sqrt{2\pi} = \frac{1}{\sqrt{2\pi}}$, solving the ODE gives

$$G(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

so $G \geq 0$ and $\int_{\mathbb{R}} G = 1$. The family $\{G_\lambda : \lambda > 0\}$ is the **Gauss kernel**.

5.2.8 Transforms of the kernels

The inversion formula says that if g and $\widehat{g}$ are continuous L^1-functions, then

$$g(x) = \frac{1}{2\pi}\widehat{\widehat{g}}(-x), \qquad x \in \mathbb{R}. \tag{17}$$

Suppose $f, g \in L^1(\mathbb{R})$, g continuous, satisfy

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} g(\alpha)e^{i\alpha x}\,d\alpha, \qquad x \in \mathbb{R}.$$

Then $f(x) = \frac{1}{2\pi}\widehat{g}(-x)$, so $\widehat{g} \in L^1(\mathbb{R})$. By the inversion formula for g (valid everywhere, g being continuous),

$$g(y) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{g}(\alpha)e^{i\alpha y}\,d\alpha = \int_{\mathbb{R}} f(-\alpha)e^{i\alpha y}\,d\alpha = \int_{\mathbb{R}} f(\beta)e^{-i\beta y}\,d\beta = \widehat{f}(y).$$

So $\widehat{f} = g$. Applying this to the three kernels:

$$\widehat{K_\lambda}(\alpha) = \max\Big\{1 - \frac{|\alpha|}{\lambda}, 0\Big\}, \qquad \widehat{P_\lambda}(\alpha) = e^{-|\alpha/\lambda|}, \qquad \widehat{G_\lambda}(\alpha) = e^{-\alpha^2/2\lambda^2},$$

where $P_\lambda = H_\lambda$ is the Poisson kernel.

Exercise 5.2.6

For $f \in L^1(\mathbb{R})$ express $\widehat{K_\lambda * f}$ in terms of $\widehat{f}$. Use this to show that $\{f \in L^1(\mathbb{R}) : \widehat{f} \in C_c(\mathbb{R})\}$ is dense in $L^1(\mathbb{R})$, and compare with Theorem 5.1.7 for Fourier series.

Solution [Supplement]**.** By Proposition 5.2.1,

$$\widehat{K_\lambda * f}(\alpha) = \widehat{K_\lambda}(\alpha)\widehat{f}(\alpha) = \max\Big\{1 - \frac{|\alpha|}{\lambda}, 0\Big\}\widehat{f}(\alpha),$$

continuous and zero outside $[-\lambda, \lambda]$, so $K_\lambda * f$ lies in the set. As $K_\lambda(x) = \lambda K(\lambda x)$, $K \geq 0$, $\int K = 1$, (10) holds and Theorem 5.2.4 gives $\|K_\lambda * f - f\|_1 \to 0$. On $\mathbb{T}$, $\widehat{K_n * f}(k) = (1 - \frac{|k|}{n+1})\widehat{f}(k)$ vanishes for $|k| > n$: compactly supported transforms on $\mathbb{R}$ correspond to trigonometric polynomials on $\mathbb{T}$, and both density results use the Fejér kernel. □

5.2.9 The Fourier transform as a homomorphism

$f \mapsto \widehat{f}$ is an injective linear map $L^1(\mathbb{R}) \to C_0(\mathbb{R})$. For $f \in C_c^2(\mathbb{R})$, Proposition 5.2.2(b) gives $\widehat{f''} = -\alpha^2\widehat{f}$, so $|\widehat{f}(\alpha)| \leq \min\{\|f\|_1, \|f''\|_1/\alpha^2\}$ and $\widehat{f} \in L^1(\mathbb{R})$; by (17), $f = \widehat{k}$ with $k(\alpha) = \frac{1}{2\pi}\widehat{f}(-\alpha)$. So $C_c^2(\mathbb{R}) \supset C_c^\infty(\mathbb{R})$ lies in the range and is dense in $C_0(\mathbb{R})$ (Proposition 5.2.5):

Theorem 5.2.8

The Fourier transform $f \mapsto \widehat{f}$ is an injective, norm-decreasing homomorphism from the Banach algebra $L^1(\mathbb{R})$ into $C_0(\mathbb{R})$, and its range is dense.

Here $C_0(\mathbb{R})$ has the pointwise product. $f \mapsto \widehat{f} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ (Theorem 5.1.11) is also a Fourier transform in this sense.

Exercise 5.2.7

Show that the range of $f \mapsto \widehat{f} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is dense.

Solution [Supplement]**.** Each finitely supported a is the transform of the trigonometric polynomial $\sum_n a(n)u_n$, and for $a \in c_0(\mathbb{Z})$, $a_N = a\chi_{[-N,N]}$ has $\|a - a_N\|_\infty = \sup_{|n|>N}|a(n)| \to 0$: the range is dense. □

5.2.10 Poisson summation

How are Fourier series and Fourier integrals related? Given $f \in L^1(\mathbb{R})$, as in Fact B.12 the series

$$F(t) = \sum_{n=-\infty}^{\infty} f(t - 2n\pi) \tag{18}$$

converges absolutely a.e., and $F \in L^1(\mathbb{T})$ ($\int_{\mathbb{T}} \sum_n |f(t-2n\pi)|\,dt = \frac{1}{2\pi}\|f\|_1$). By Levi's theorem,

$$\widehat{F}(n) = \int_{\mathbb{T}} F(t)e^{-int}\,dt = \sum_{k=-\infty}^{\infty} \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t+2k\pi)e^{-int}\,dt = \frac{1}{2\pi}\int_{\mathbb{R}} f(t)e^{-int}\,dt = \frac{1}{2\pi}\widehat{f}(n). \tag{19}$$

Hence if the Fourier series of F converges at $t \in \mathbb{T}$ (to $F(t)$), we get the **Poisson summation formula**

$$\sum_{n=-\infty}^{\infty} f(t - 2n\pi) = \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} \widehat{f}(n)e^{int}.$$

Exercise 5.2.8

Use $f(x) = e^{-|x|}$ to prove

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2+1} = \pi\,\frac{e^{2\pi}+1}{e^{2\pi}-1}.$$

Solution [Supplement]**.** $\widehat{f}(\alpha) = \frac{2}{1+\alpha^2}$ (Example 1), so $\widehat{F}(n) = \frac{1}{\pi}\frac{1}{1+n^2} \in \ell^1(\mathbb{Z})$. $F(t) = \sum_n e^{-|t-2n\pi|}$ converges uniformly on $[-\pi,\pi]$ (terms $\le e^{-(2|n|-1)\pi}$), so F is continuous, and by Corollary 5.1.9 and Exercise 5.1.1 $F(t) = \sum_n \widehat{F}(n)e^{int}$ for every t. At $t = 0$:

$$\sum_n e^{-2\pi|n|} = \frac{1}{\pi}\sum_n \frac{1}{1+n^2}.$$

The left side is $1 + \frac{2e^{-2\pi}}{1-e^{-2\pi}} = \frac{1+e^{-2\pi}}{1-e^{-2\pi}} = \frac{e^{2\pi}+1}{e^{2\pi}-1}$, which gives the formula. □

Exercise 5.2.9

For $f \in L^1(\mathbb{R})$ and $\lambda > 0$ define, as in (18),

$$F_\lambda(t) = \sum_{n=-\infty}^{\infty} f_\lambda(t - 2n\pi), \qquad f_\lambda(x) = \lambda f(\lambda x).$$

Show that

$$\widehat{F_\lambda}(n) = \frac{1}{2\pi}\widehat{f}\Big(\frac{n}{\lambda}\Big), \qquad \lim_{\lambda\to\infty}\|F_\lambda\|_{L^1(\mathbb{T})} = \frac{1}{2\pi}\|f\|_{L^1(\mathbb{R})}.$$

Use this to prove Theorem 5.2.7.

Solution [Supplement]**.** By (19) and Exercise 5.2.4, $\widehat{F_\lambda}(n) = \frac{1}{2\pi}\widehat{f_\lambda}(n) = \frac{1}{2\pi}\widehat{f}(n/\lambda)$. Upper bound: $\|F_\lambda\|_{L^1(\mathbb{T})} \le \frac{1}{2\pi}\sum_n \int_{-\pi}^{\pi} |f_\lambda(t-2n\pi)|\,dt = \frac{1}{2\pi}\|f_\lambda\|_1 = \frac{1}{2\pi}\|f\|_1$. Lower bound: separating the $n=0$ term,

$$\|F_\lambda\|_{L^1(\mathbb{T})} \ge \frac{1}{2\pi}\Big[\int_{-\pi}^{\pi}|f_\lambda(t)|\,dt - \sum_{n\neq 0}\int_{-\pi}^{\pi}|f_\lambda(t-2n\pi)|\,dt\Big] = \frac{1}{2\pi}\Big[\int_{|x|\le\pi}|f_\lambda| - \int_{|x|>\pi}|f_\lambda|\Big]$$
$$= \frac{1}{2\pi}\Big[\int_{|y|\le\lambda\pi}|f| - \int_{|y|>\lambda\pi}|f|\Big] \longrightarrow \frac{1}{2\pi}\|f\|_1 \quad (\lambda\to\infty).$$

Proof of Theorem 5.2.7: if $\widehat{f}=0$, then $\widehat{F_\lambda}(n)=0$ for all n and λ, so $F_\lambda = 0$ a.e. by Theorem 5.1.8, i.e. $\|F_\lambda\|_{L^1(\mathbb{T})}=0$. Letting $\lambda\to\infty$ gives $\|f\|_1 = 0$. □